QUADRATIC EQUATIONS — SOLVING

Digital SAT Style | Total: 60 Questions | Mixed Difficulty

🟢 *Easy: Q1–Q12* 🟡 *Medium: Q13–Q48* 🔴 *Hard: Q30–Q60*

Section A — Solving by Factoring (Q1–Q12)

Q1. 🟢 **Easy**

Which of the following is a solution to the equation $x^2 - 7x + 12 = 0$?

A) $x = 2$
B) $x = 3$
C) $x = 5$
D) $x = 6$

Q2. 🟢 **Easy**

What are the solutions to $x^2 - x - 6 = 0$?

A) $x = -2$ and $x = 3$
B) $x = 2$ and $x = -3$
C) $x = -1$ and $x = 6$
D) $x = 1$ and $x = -6$

Q3. 🟢 **Easy**

Which value of x satisfies $2x^2 + 5x - 3 = 0$?

A) $x = -3$
B) $x = 1/2$
C) $x = 3$
D) $x = -1/2$

Q4. 🟢 **Easy**

The equation $x^2 + 6x + 9 = 0$ has how many distinct real solutions?

A) 0
B) 1
C) 2
D) 3

Q5. 🟢 **Easy**

Which of the following correctly factors $x^2 - 16$?

A) $(x - 4)^2$
B) $(x + 4)(x - 4)$
C) $(x - 8)(x + 2)$
D) $(x + 4)^2$

Q6. 🟢 **Easy**

If $(x - 5)(x + 2) = 0$, what are all values of x?

A) $x = 5$ only
B) $x = -2$ only
C) $x = 5$ and $x = -2$
D) $x = -5$ and $x = 2$

Q7. 🟢 **Easy**

What is the positive solution to $x^2 - 9x + 18 = 0$?

GRID-IN *(Enter your answer in the box provided)*

Q8. 🟢 **Easy**

Which of the following is a factor of $3x^2 + 10x - 8$?

A) $(3x - 2)$
B) $(x + 4)$
C) $(3x + 2)$
D) $(x - 4)$

Q9. 🟢 **Easy**

What is the sum of the solutions to $x^2 - 8x + 12 = 0$?

A) 4
B) 6
C) 8
D) 12

Q10. 🟢 **Easy**

If $x^2 = 49$, what are all values of x?

A) $x = 7$ only
B) $x = -7$ only
C) $x = 7$ and $x = -7$
D) $x = 7$ and $x = 0$

Q11. 🟢 **Easy**

Which equation has $x = -4$ and $x = 1$ as its solutions?

A) $x^2 + 5x - 4 = 0$
B) $x^2 - 5x - 4 = 0$
C) $x^2 + 3x - 4 = 0$
D) $x^2 - 3x + 4 = 0$

Q12. 🟢 **Easy**

What is the product of the solutions to $2x^2 - 7x - 4 = 0$?

A) -2

B) 2
C) −4
D) 7/2

Section B — Quadratic Formula & Discriminant (Q13–Q24)

Q13. 🟡 **Medium**

Using the quadratic formula, what are the solutions to $x^2 - 4x - 1 = 0$?

A) $x = 2 \pm \sqrt{5}$
B) $x = -2 \pm \sqrt{5}$
C) $x = 2 \pm \sqrt{3}$
D) $x = 4 \pm \sqrt{5}$

Q14. 🟡 **Medium**

How many real solutions does the equation $2x^2 - 3x + 5 = 0$ have?

A) 0
B) 1
C) 2
D) Cannot be determined

Q15. 🟡 **Medium**

The discriminant of $3x^2 + bx + 12 = 0$ equals zero. What is the positive value of b?

A) 6
B) 9
C) 12
D) 18

Q16. 🟡 **Medium**

In the given equation, c is a constant. The equation $4x^2 + 12x + c = 0$ has exactly one real solution. What is the value of c?

A) 3
B) 6
C) 9
D) 12

Q17. 🟡 **Medium**

What is the larger of the two solutions to $x^2 - 6x - 7 = 0$?

A) 1
B) 6
C) 7
D) −1

Q18. 🟡 **Medium**

What is the positive solution of $5x^2 - 3x - 2 = 0$?

GRID-IN *(Enter your answer in the box provided)*

Q19. 🟡 **Medium**

For which values of k does $x^2 + kx + 25 = 0$ have exactly two distinct real solutions?

A) $k > 10$
B) $k < -10$ or $k > 10$
C) $-10 < k < 10$
D) $k = 10$ only

Q20. 🟡 **Medium**

Which of the following is equivalent to $x^2 - 10x + 22$ after completing the square?

A) $(x - 5)^2 - 3$
B) $(x - 5)^2 + 3$
C) $(x + 5)^2 - 3$
D) $(x - 10)^2 + 22$

Q21. 🟡 **Medium**

The solutions to $ax^2 + bx + c = 0$ are $x = 1/3$ and $x = -5$. What is the value of b/a?

A) $-14/3$
B) $14/3$
C) $-4/3$
D) $4/3$

Q22. 🟡 **Medium**

If $3x^2 - 7x + k = 0$ has one solution of $x = 1/3$, what is the value of k?

A) -2
B) 2
C) 6
D) -6

Q23. 🟡 **Medium**

Which of the following is a solution to $x^2 + 2x - 8 = 0$ that is also negative?

A) $x = -4$
B) $x = -2$
C) $x = -6$
D) $x = -1$

Q24. 🟡 **Medium**

What is the sum of the two solutions to $2x^2 - 5x - 3 = 0$?

GRID-IN *(Enter your answer in the box provided)*

Section C — Vertex Form & Completing the Square (Q25–Q36)

Q25. 🟡 Medium

The function $f(x) = (x - 3)^2 + 7$ has a minimum value at $x =$?

A) 3
B) −3
C) 7
D) −7

Q26. 🟡 Medium

Which of the following gives the vertex of the parabola $y = 2x^2 - 8x + 5$?

A) (2, −3)
B) (−2, 25)
C) (4, 5)
D) (2, 5)

Q27. 🟡 Medium

The graph of $y = -x^2 + 6x - 5$ has its maximum at the point:

A) (3, 4)
B) (3, −4)
C) (6, −5)
D) (−3, 4)

Q28. 🟡 Medium

A parabola has vertex (−2, 5) and passes through (0, 1). Which equation models this parabola?

A) $y = -(x + 2)^2 + 5$
B) $y = (x + 2)^2 + 5$
C) $y = -(x - 2)^2 + 5$
D) $y = (x - 2)^2 - 5$

Q29. 🟡 Medium

What is the axis of symmetry of the parabola $f(x) = 3x^2 + 12x - 4$?

A) $x = -2$
B) $x = 2$
C) $x = -4$
D) $x = 4$

Q30. 🔴 Hard

The graph of $y = a(x - h)^2 + k$ passes through (1, 0) and has vertex (3, 8). What is the value of a?

A) −2
B) 2
C) −8
D) 8

Q31. 🔴 **Hard**

The function $g(x) = x^2 + bx + 16$ has a minimum value of 7. What is the positive value of b?

A) 3
B) 6
C) 9
D) 12

Q32. 🔴 **Hard**

A quadratic function has zeros at $x = -1$ and $x = 5$, and passes through $(0, -10)$. What is the leading coefficient?

GRID-IN *(Enter your answer in the box provided)*

Q33. 🔴 **Hard**

The y-intercept of the parabola with vertex $(4, -3)$ and containing the point $(2, 1)$ is:

A) (0, 1)
B) (0, 13)
C) (0, 5)
D) (0, −3)

Q34. 🔴 **Hard**

Function f is quadratic with vertex at $(-2, 9)$ and a y-intercept at $(0, 5)$. Which of the following gives $f(x)$?

A) $f(x) = -(x + 2)^2 + 9$
B) $f(x) = -(x - 2)^2 + 9$
C) $f(x) = (x + 2)^2 + 9$
D) $f(x) = -2(x + 2)^2 + 9$

Q35. 🔴 **Hard**

If $f(x) = ax^2 + bx + c$ has a minimum at $(3, -4)$ and $f(0) = 5$, which of the following must be true?

A) $a > 0$ and $c = 5$
B) $a < 0$ and $c = 5$
C) $a > 0$ and $c = -4$
D) $a > 0$ and $c = 3$

Q36. 🔴 **Hard**

A quadratic has a maximum value of 16 at $x = 2$. Which of the following could be the equation?

A) $y = -2(x - 2)^2 + 16$
B) $y = 2(x - 2)^2 + 16$
C) $y = -2(x + 2)^2 + 16$

D) $y = 2(x + 2)^2 - 16$

Section D — Word Problems (Q37–Q48)

Q37. 🟡 **Medium**

A ball is thrown upward. Its height h, in feet, t seconds after launch is $h(t) = -16t^2 + 64t + 5$. What is the maximum height, in feet, the ball reaches?

A) 64
B) 69
C) 80
D) 5

Q38. 🟡 **Medium**

The area of a rectangular garden is 84 square feet. The length is 2 feet more than twice the width. What is the width, in feet?

A) 6
B) 8
C) 10
D) 12

Q39. 🟡 **Medium**

A company's profit P, in thousands of dollars, after selling x hundred units is $P(x) = -x^2 + 8x - 7$. How many hundred units must be sold to break even ($P = 0$)?

A) 1 and 7
B) 2 and 4
C) 1 and 8
D) 3 and 5

Q40. 🟡 **Medium**

The product of two consecutive even integers is 288. What is the smaller of the two integers?

GRID-IN *(Enter your answer in the box provided)*

Q41. 🟡 **Medium**

A rectangular picture frame has a perimeter of 40 inches and an area of 84 square inches. What is the length, in inches, of the longer side?

A) 12
B) 14
C) 16
D) 7

Q42. 🔴 **Hard**

A ball is launched from a platform 28 feet high with initial velocity 48 ft/s. Its height is $h(t) = -16t^2 + 48t + 28$. After how many seconds does the ball hit the ground?

A) 2
B) 3
C) 3.5
D) 4

Q43. 🔴 **Hard**

The revenue R, in dollars, from selling x items is $R(x) = 80x - x^2$. The cost is $C(x) = 20x + 400$. At what value of x is profit maximized?

A) $x = 20$
B) $x = 30$
C) $x = 40$
D) $x = 60$

Q44. 🔴 **Hard**

Two positive numbers differ by 4. The sum of their squares is 106. What is the smaller number?

GRID-IN *(Enter your answer in the box provided)*

Q45. 🔴 **Hard**

A parabolic arch has height 18 feet at its peak, which is 12 feet from either end. A truck 8 feet wide needs to pass under the arch centered below the peak. What is the maximum height, in feet, available at the truck's outer edge (4 feet from center)?

A) 14
B) 15
C) 12
D) 16

Q46. 🔴 **Hard**

A rock is thrown from the edge of a cliff. Its height h in meters after t seconds is $h(t) = -5t^2 + 20t + 60$. At what time does the rock reach the same height as the cliff ($h = 60$)?

A) $t = 0$ and $t = 4$
B) $t = 2$ only
C) $t = 4$ only
D) $t = 3$ and $t = 5$

Q47. 🔴 **Hard**

A square has its side length increased by 5. The new area is 169 square units. What was the original side length?

GRID-IN *(Enter your answer in the box provided)*

Q48. 🔴 **Hard**

The path of a kicked soccer ball follows $h(x) = -0.04x^2 + 1.6x$ where x is horizontal distance (feet) and h is height (feet). What is the maximum height reached?

A) 12 feet
B) 14 feet
C) 16 feet
D) 20 feet

Section E — Systems, Intersections & Hard Mixed (Q49–Q60)

Q49. 🔴 **Hard**

At how many points does $y = x^2 - 5x + 6$ intersect $y = x - 1$?

A) 0
B) 1
C) 2
D) 3

Q50. 🔴 **Hard**

A line $y = 2x + k$ is tangent to the parabola $y = x^2 + 1$. What is the value of k?

A) −2
B) 0
C) −1
D) 1

Q51. 🔴 **Hard**

If $x^2 - 6x + p = 0$ has two equal real roots, what is the value of p?

A) 3
B) 6
C) 9
D) 12

Q52. 🔴 **Hard**

The parabola $y = x^2 + bx + c$ has its vertex at $(3, -4)$. What is the value of c?

GRID-IN *(Enter your answer in the box provided)*

Q53. 🔴 **Hard**

A quadratic equation has roots r and s such that $r + s = 7$ and $rs = 10$. What is the equation?

A) $x^2 - 7x + 10 = 0$
B) $x^2 + 7x - 10 = 0$
C) $x^2 - 7x - 10 = 0$
D) $x^2 + 7x + 10 = 0$

Q54. 🔴 **Hard**

For what values of m does the system $y = x^2$ and $y = mx - 4$ have exactly one solution?

A) $m = \pm 4$
B) $m = \pm 2$
C) $m = \pm 8$
D) $m = 0$ only

Q55. 🔴 **Hard**

If $f(x) = x^2 - 4$ and $g(x) = x + 2$, what are all x-values where $f(x) = g(x)$?

A) $x = 3$ and $x = -2$
B) $x = 3$ and $x = -1$
C) $x = 2$ and $x = -3$
D) $x = 6$ and $x = -1$

Q56. 🔴 **Hard**

The function $f(x) = x^2 - 2x - 8$ crosses the x-axis at two points. What is the distance between these two x-intercepts?

A) 4
B) 5
C) 6
D) 8

Q57. 🔴 **Hard**

The sum of a number and its square is 56. What is the positive value of that number?

GRID-IN *(Enter your answer in the box provided)*

Q58. 🔴 **Hard**

If the graph of $y = x^2 + bx + c$ is symmetric about $x = 4$ and passes through $(7, 6)$, which of the following represents the equation?

A) $y = (x - 4)^2 - 3$
B) $y = (x + 4)^2 - 3$
C) $y = (x - 4)^2 + 3$
D) $y = (x - 7)^2 + 6$

Q59. 🔴 **Hard**

When the quadratic expression $2x^2 + px + q$ is divided by $(x - 3)$, the remainder is -6. When divided by $(x + 1)$, the remainder is 6. What is the value of $p + q$?

A) -10
B) -8
C) 2
D) 8

Q60. 🔴 **Hard**

A quadratic function f satisfies $f(1) = 0$, $f(-3) = 0$, and $f(0) = -6$. Which of the following defines f?

A) $f(x) = 2(x - 1)(x + 3)$

B) $f(x) = -2(x - 1)(x + 3)$
C) $f(x) = 2(x + 1)(x - 3)$
D) $f(x) = (x - 1)(x + 3)$